Instruction Pipelining Principles

Q. What is Instruction Pipelining? Explain the stages of a standard instruction pipeline, its working mechanism, and its advantages.

> **Definition to Remember**

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1. Stages of a Standard Pipeline

A classic CPU pipeline divides instruction execution into four distinct stages:

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2. Working Mechanism

- **Without Pipelining (Sequential):** The CPU finishes all 4 stages of Instruction 1 before starting Instruction 2. (3 instructions take 12 clock cycles).

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3. Pipeline Timing Diagram

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Clock

## 4. Advantages of Pipelining

- **Increased Throughput:** Drastically increases the number of instructions executed per second. (Ideal CPI approaches 1).

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> **Must-Write Points (for 10 marks)**

> 1. **
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> **Quick Recall**

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